

Human Adaptation in Creative Process with Generative AI*

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Version: August 2026

Abstract

What is the role of human input in AI-assisted production? Humans interact with generative AI through combinations of words called prompts. Users dynamically modify their prompts based on the AI’s previous output. I empirically examine how users adjust their prompts iteratively to converge on desired outcomes. I study this creative process using prompt-level data from Midjourney, a leading AI image generator. I document novel facts about prompt patterns in image production and find that they are similar to consumer product searching patterns. (1) Prompts are path-dependent; (2) Prompt length increases step-by-step; (3) Prompts converge to the final prompt along the paths; (4) Users adjust words with greater weights first and then move to less important words. I estimate a structural model of the creative process using the sequential search framework. Counterfactual shows that without human learning in the loop, users need three times more prompts to achieve data-observed results.

*I am grateful to my advisors, Matthew Mitchell, Heski Bar-Isaac, Avi Goldfarb, and Ambarish Chandra, for their guidance and support. I also thank Charles Hodgson, Victor Aguirregabiria, Yanyou Chen, Yao Luo, Siyuan Liu, Irisa Zhou, Ruiqi Sun, Hong Luo, Kevin Bryan, Stephan Heblich, Tommaso Alba, Devin Bissky Dziadyk, Ben Couillard, and the participants in the IO seminar at the University of Toronto for their helpful comments. I thank Ali Berk Senbas for his insights. I gratefully acknowledge support by the Deutsche Forschungsgemeinschaft (DFG) through CRC TR 224 (project B05). This paper was previously circulated under the title “Learning to Prompt: Human Adaptation in Production with Generative AI.” All errors are my own.

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1 Introduction

Generative AI has rapidly expanded into creative domains such as creative writing, drafting emails, and producing visual art. These are domains that were previously thought to require distinctly human effort. Accordingly, creative industries are incorporating generative AI into their workflows. For example, AI companies have formed partnerships with film, gaming, and news studios to generate illustrations and animated characters.¹ This leads to a public discussion among creative workers: What is the role of human input in a creative process assisted by AI? For instance, with AI being able to generate an artistic image within seconds, what is the human effort in this creative process?

In an AI-assisted creative process, initial outputs from AI rarely meet users' requirements and need further adjustments. Users typically refine prompts iteratively based on previous results to converge towards desired outcomes. This process suggests that human input does not disappear with generative AI. Rather, it shifts from directly producing the image to guiding, evaluating, and revising the production process. Professional illustrator Christoph Niemann writes in the New York Times Magazine: "Whether you are a human or a machine, creative work consists of a thousand tiny decisions. But how does one decide if an element is good, funny, say, witty or absurd?"² His description and accompanying illustrations highlight this iterative decision-making process. In this paper, I argue that the role of human input in an AI-assisted creative process lies in iteratively refining prompts through human judgment and adaptation, and I quantify their impact on the creative process in a measurable way.

The empirical setting is based on Midjourney, a popular AI image generator that operates through public chat channels on Discord, a messaging platform. In this environment, users submit prompts, which are a combination of words, in the chat

¹<https://www.theguardian.com/film/2024/sep/18/lionsgate-ai>; <https://developer.microsoft.com/en-us/games/articles/2023/11/xbox-and-inworld-ai-partnership-announcement>; <https://www.reuters.com/technology/financial-times-openai-sign-content-licensing-partnership-2024-04-29>.

²<https://www.nytimes.com/interactive/2025/06/23/magazine/ai-art-artists-illustrator.html>

box to the AI, which then returns generated images in exchange for a fee. This setting allows me to observe both submitted prompts and generated images over time, along with the user ID. Both anecdotal evidence and patterns in the data show that users adjust their prompts iteratively as they move toward desired outcomes. This setting, therefore, provides a useful environment for studying how human judgment shapes production with generative AI. Using reduced-form analysis, I examine how users revise prompts in response to images generated from earlier prompts.

I document four patterns in prompt construction within a session. First, prompts are path dependent: later prompts build on earlier ones rather than being written independently. Second, prompt length increases incrementally over time. Third, users prioritize adjusting words with higher weights before addressing less influential words. Fourth, prompts become increasingly similar to the final prompt in the session. Together with anecdotal evidence, these patterns suggest that users iteratively evaluate AI-generated images, revise their prompts, and converge toward their desired outcomes.

Motivated by these data patterns, I develop a structural model of the creative process using the sequential search framework from Weitzman (1979) and Hodgson and Lewis (2025). In the model, individuals choose a prompt to submit to the AI based on their current beliefs about the expected image quality of prompts. They are uncertain about the generated image quality until they observe the image. Observing an additional image requires paying a search cost, which includes the mental effort required to construct prompts and judge image quality, the fees paid to the AI company, and the waiting time. They then update their beliefs about the expected returns of prompts in their consideration set and either stop or choose the next prompt accordingly. The consideration set is defined as a local set of prompts similar in content to others within the same session, representing ideas that belong to the same artwork. When a prompt is submitted to the AI and the individual observes a less satisfying image, they infer that similar prompts are more likely to produce unsatisfying results. These prompts will be chosen with lower probability in the next search. Individuals stop searching when no remaining prompts are expected to generate sufficient value to compensate for the search cost. The trade-off is between

the search cost and the option value of the next prompt.

A counterfactual analysis investigates how removing adaptation from previous prompt outputs affects the prompt construction process. In the counterfactual scenario, users can only submit a list of prompts to the AI at the start of a session. They do not adjust the order of the prompts in the list when they see the generated image sequentially. The results show that eliminating judgment on image quality increases the number of prompts per creative process by 313% to achieve the same outcome as in the data, implying that adaptive learning plays a crucial role in guiding efficient prompt searching behavior. This finding quantifies the critical role of human adaptation: the ability to learn from prior outputs and iteratively guide the AI makes the creative process more efficient.

In summary, I investigate the creative process of using generative AI to produce digital art. A key contribution of this paper is to empirically document patterns in this process, showing that it closely aligns with search and learning. In previous literature on creative works, for example, in scientific discovery, researchers usually focus on the inputs (funding from government, number of scientists, etc.) and outputs (patent citations, etc.), while the creative process itself remains a black box. This context provides a unique setting to open this black box by observing the inputs and outputs of *intermediate steps* of the creative process. I show that human adaptation during the process guides the direction of creation and enhances production efficiency. Finally, this is among the earliest studies to structurally model the creative process as a costly sequential search process, providing an infrastructure for future researchers to study the production process of creative works.

Related Literature

This paper connects to two strands of literature. First, it contributes to the recent literature that analyzes conversational data between humans and generative AI (Chatterji et al., 2025; Handa et al., 2025). This literature focuses on the purpose of using generative AI. Handa et al. (2025) shows that AI usage mostly concentrates on coding and writing tasks. Chatterji et al., 2025 finds that users mostly use generative AI to obtain information and improve writing.

While I also analyze conversational data, this paper focuses on how users interact with AI. In Section 4, I document four novel patterns about AI usage in the creative process. Motivated by these patterns, I structurally model the creative process using the costly sequential search framework in Section 6.

Second, this paper is related to the sequential search model literature (Weitzman, 1979; Kim, Albuquerque, and Bronnenberg, 2010; Bronnenberg, Kim, and Mela, 2016; Chen and Yao, 2017; De los Santos and Koulayev, 2017; De Los Santos, Hortaçsu, and Wildenbeest, 2017; Dzyabura and Hauser, 2019; Honka, Hortaçsu, and Wildenbeest, 2019; Hirsch, Guy, Nus, Dagan, and Kurland, 2020; Ursu, Wang, and Chintagunta, 2020; Hodgson and Lewis, 2025) which provides the foundation for the structural model in Section 6. Weitzman (1979) propose the theoretical framework of the sequential search model, in which individuals search sequentially across potential research projects. Individuals prioritize projects with the highest expected returns and gradually proceed down the ranked list until no remaining project offers sufficient expected returns to justify the search cost. This framework has been widely adopted in consumer search contexts, where consumers search sequentially for products to purchase (Kim et al., 2010; Chen and Yao, 2017; De los Santos and Koulayev, 2017). These papers estimate search cost distributions using consumers’ browsing history data. While earlier work assumes static beliefs about unsearched products, more recent studies incorporate learning into consumer search models, allowing beliefs to update along the search path (De Los Santos et al., 2017; Dzyabura and Hauser, 2019; Ursu et al., 2020; Hodgson and Lewis, 2025). For example, a consumer encountering negative reviews for one camera of a certain brand may update their beliefs about all cameras of this brand. The most closely related work is Hodgson and Lewis (2025), in which consumers update their beliefs about camera quality during the search process. Upon observing a satisfactory camera, consumers infer that similar cameras are also likely to be satisfactory, increasing the probability of searching related products in subsequent periods.

I adapt this framework to the creative context by interpreting the image generation process as a search for the optimal prompt. In the structural model described in Section 6, individuals hold prior beliefs about prompt quality based on prompt

attributes. They pay a search cost to observe and evaluate the generated image, then update their beliefs about prompts in their consideration set. This AI feedback informs them about the next prompt to submit to the AI. The counterfactual exercise suggests that this feedback-driven learning is essential for achieving production efficiency.

The remainder of the paper is structured as follows. I introduce the context and data in Section 2 and Section 3. In Section 4, I present the data patterns in a single creative process. Section 6 describes the model setup. Section 8 runs a counterfactual simulation of the structural model. Section 9 concludes the paper.

2 Context

The context studied in this paper is Midjourney. It has been one of the most popular AI text-to-image generators during the observation period. Midjourney enables users to generate digital images with its AI models by submitting prompts, typically text and sometimes image references, in exchange for a subscription fee. Before August 2024, Midjourney was only accessible on a public chat channel on Discord, a messaging service widely used by game streamers and communities for real-time interaction. Although Midjourney introduced its own web-based interface for all users in August 2024, a substantial portion of its user base has continued to rely on Discord to access the service.

The partnership between Midjourney and Discord is driven by two primary considerations. Anecdotally, during its early beta phase, Midjourney operated as a small startup with limited financial resources, and building on top of Discord’s existing infrastructure offered a cost-effective way to provide image generation services. Additionally, in an interview, Midjourney’s founder and CEO, David Holz, explained that a central motivation was to cultivate an interactive community where users could openly discuss ideas, share inspiration, and collaborate on refining prompt-crafting strategies.³

³<https://www.theverge.com/2022/8/2/23287173/ai-image-generation-art-midjourney->

A key advantage of this empirical setting is that it permits direct observation of human interaction with generative AI. This feature allows me to analyze how users iteratively adjust their prompts and gradually converge to their desired outcomes. In many other creative settings, researchers only observe the input of the creative process, such as the monetary investment, the number of scientists and their characteristics, and the output of the creative process, such as patents or journal publications. The creative process itself remains unobservable. However, in this setting, a significant part of the creative process can be directly observed. This allows me to study the role of human adaptation in a single creative process of an artwork.

Additionally, Midjourney is both influential and representative. Although the company has not released official figures on its market share, independent listings and roundups consistently rank Midjourney among the most popular AI image generators, alongside DALL-E, Stable Diffusion, Adobe Firefly, and Leonardo AI.⁴ By October 2025, Midjourney has become the largest Discord server by number of users.⁵

It is helpful to characterize Midjourney’s user base and understand the purposes for which the platform is used. Midjourney does not publish official statistics on the composition of its users, but several alternative sources provide relevant information. Within Midjourney’s Discord server, the “micro-polls” and the “discussion” sub-channel contain user surveys and discussions.⁶ Users also discuss Midjourney on social media platforms such as Reddit, Twitter, and Instagram. In addition, SimilarWeb, a web analytics company, reports the other websites that Midjourney users frequently visit. Taken together, these sources provide useful evidence about the characteristics of Midjourney users. Some screenshots of the “micro-polls” and the SimilarWeb report can be found in Appendix A.

Many Midjourney users appear to have experience in art and design. In self-reported discussions and polls, users indicate that they currently work, or have pre-

multiverse-interview-david-holz

⁴<https://www.demandsage.com/ai-image-generators/>; <https://www.cmswire.com/digital-marketing/a-marketers-guide-to-the-best-text-to-image-generators/>; <https://www.forbes.com/lists/ai50/>

⁵<https://backlinko.com/discord-users>; <https://influencermarketinghub.com/discord-stats/>

⁶These polls are usually initiated by Midjourney’s CEO David Holz. According to the description of this sub-channel, Midjourney is using it to construct larger questionnaires, which he later publishes in the “announcements” sub-channel for users to fill out.

viously worked, in art or design industries. Some also report owning professional drawing tablets such as Wacom devices and subscribing to software such as Adobe Photoshop. Their online browsing patterns further suggest an interest in creative production: in addition to Midjourney, they frequently visit other AI services such as ChatGPT, graphic-design platforms, art-resource websites such as Behance, marketplaces for selling customized products such as T-shirts and mugs, and freelance platforms such as Fiverr. The user base therefore includes both hobbyists and professional creators. It also extends beyond visual artists to other types of creators, including fiction writers and musicians, as well as some small-business owners.

Users turn to Midjourney for a wide range of purposes. Many of them are using Midjourney purely for fun. Others use it to produce decorative content for social media, wallpapers, or products such as T-shirts, which may sometimes be sold commercially. Some artists incorporate Midjourney into their creative workflow by using it to explore alternative color schemes, compositions, and visual concepts before developing the final work on a physical canvas. Midjourney is also used for commercial applications, including book illustrations, album covers, and logo design. In addition, some users rely on the tool for creative inspiration. For instance, fiction writers may generate images to visualize settings or scenes, while musicians may use generated visuals as inspiration for their compositions.

An important feature of Midjourney is that, from versions 3 through 6, its official documentation does not describe any form of session-level memory. In particular, there is no indication that prompts submitted earlier in a user’s interaction affect the images generated from later prompts. Unlike session-based systems such as ChatGPT, Midjourney does not seem to use prior conversations as persistent context. If users want to build on an earlier output, they must explicitly provide that output again. For example, they can include a previous image in a new prompt, or copy and paste an earlier prompt, then resubmit it. In both cases, each prompt is treated as an independent input without persistent session memory. Personalization features were not introduced until version 7.

3 Data

The data contains all publicly observable messages posted in Midjourney’s Discord server between June 2022 and January 2025. User activity is recorded through messages posted by the Midjourney Bot. Each message corresponds to one action, including submitting a prompt to Midjourney or clicking a button below the generated images. For each action, I can observe the timestamp, user ID, submitted prompt, generated images, and the buttons clicked, if any. The analysis in this paper focuses primarily on prompts submitted to Midjourney.

One caveat is that users with higher-tier subscriptions can enable “stealth mode,” which prevents their activity from appearing in the public Discord channels. This selection could matter if users systematically choose stealth mode because they are concerned that publicly sharing prompts makes them easy to imitate. Discussions on Reddit and similar forums seem to suggest that this is not generally considered a serious problem. Since there are millions of users on this server and there is a high volume of messages every minute, individual messages are quickly drowned out by hundreds of others. This high volume of messages mitigates such concerns.

I focus on a random sample of 500 users who submitted at least 50 prompts in each of V3, V4, V5, and V6 in the dataset, along with their complete usage history. There are many casual users in this public chat channel who submitted a few prompts out of curiosity and then quit. By focusing on these frequent users, I constrain the attention to the more serious users. To construct the consideration set, the sample is restricted to a single genre, landscapes, pooling all sessions from the 500 focal users and grouping them by topic.

An illustrative example of a prompt is as follows.

```
**<https://s.mj.run/ABCDEFG> dark fantasy, retro anime style, mink  
hiding in a forest, horror core, ultra realistic, retro, video game style, creepy -  
- aspect 5:4** - <@123456789987654321>
```

```
07/16/2023 11:18:25.796 UTC
```

In this example, “https://s.mj.run/ABCDEFGF” corresponds to an image input; “dark fantasy, retro anime style, mink hiding in a forest, horror core, ultra realistic, retro, video game style, creepy” is the text input; “aspect 5:4” is a parameter, and “123456789987654321” is the user ID. A single prompt may contain more than one image reference and several parameters, and each submission is recorded with a timestamp. The analysis that follows uses only the textual component of each prompt.

3.1 Cluster Prompts to Define Session

An important observation from the data is that users submit sequences of correlated prompts within short time periods, generating visually similar images. The following example illustrates this behavior.

1. “a closeup of a book cover on a beach on a rainy day”
2. “closeup of a book cover lying in the sand on a beach”
3. “closeup of a book cover lying in the sand on a beach, an hourglass is also in the sand”
4. “closeup of a book cover lying in the sand on a beach, a small hourglass is also in the sand”

Users appear to refine their prompts iteratively to converge toward a desired outcome. This practice is consistent with online discussions where users describe such strategies as “I’d experiment by removing or slightly modifying descriptors to make it fit what I wanted”, “Rerun, refine, repeat, dominate” or “Start simple, only add what you need”. Many YouTube videos also teach Midjourney tips by showing the YouTubers’ workflow on Midjourney. In these videos, many users are demonstrating such iterative prompt refinement strategies.

Therefore, I treat these prompt sequences as intermediate steps in a production process and define a *session* as a sequence of highly correlated prompts submitted by a single user. The session serves as the production unit in this context.

	Mean	Median	99th Percentile	Min	Max
#sessions per user	191	114	1,408	7	4,120
#prompts per session	12	1	18	1	45,720
#words per prompt	28	18	195	0	378
#parameters per prompt	1.60	1	5	0	10
#image inputs per prompt	0.21	0	3	0	57
N_{user}			500		
$N_{session}$			95,545		
N_{prompt}			1,160,773		

Table 1: Summary Statistics

To identify such sessions, I implement the following clustering procedure. For each user, I calculate a Jaccard similarity matrix across all prompts and apply the Density-based spatial clustering of applications with noise (DBSCAN) algorithm, one of the most commonly used and cited clustering algorithms. I then verify the constructed sessions by examining the time span. Over 90% of sessions are finished within 24 hours. The details of clustering are described in Appendix B.3.

Table 1 represents the summary statistics. Both the median and minimum number of prompts per session equal one, indicating that a substantial share of sessions consist of a single prompt. In these cases, users submit one prompt, and either obtain a satisfying result or receive images that are sufficiently poor that they quit the session immediately. The distribution of session length is skewed and long-tailed. Besides, the minimum number of words per prompt is zero because, in rare cases, users submit only image inputs, sometimes with parameter specifications to Midjourney, without any text. Fifteen sessions contain more than 10,000 prompts, representing less than 0.02% of all sessions. These extreme cases likely reflect some rare automated prompting behaviors. Since the subsequent analysis is mainly performed at the session level, these outliers do not significantly affect the results.

3.2 Supplementary Data

Apart from messages on the Midjourney Discord server, I also use supplementary data from online Midjourney community classifications, including Midlibrary, Imiprompt, ShellyPalmer, and a popular collection of words on a GitHub repository.⁷ These data give me information like artist names, photography words, and style words, which are later used in prompt analysis.

4 Prompt Evolvment Patterns within Session

In online Midjourney communities, users share workflows for generating desired images, Common suggestions include “start simple, then elaborate, add one detail at a time”; “elements placed at the start of a prompt will have a more significant effect on the result than those placed at the end”; “regenerate the image with the same prompt until a satisfactory result appears”.⁸ These discussions suggest that formulating a satisfying prompt is mentally demanding and relies on iterative experimentation with the AI’s outputs.

Below are four patterns of prompts within a session.

4.1 Path Dependence of Prompts

Section 3.1 illustrates that in the data, there are sequences of prompts that are highly correlated with each other and are submitted to AI within a short time frame. The step-by-step strategy for adjusting the prompts has also been widely discussed on

⁷<https://midlibrary.io/>;
<https://www.imiprompt.com/>;
<https://shellypalmer.com/midjourney-reference-art/>;
<https://github.com/willwulfken/MidJourney-Styles-and-Keywords-Reference/blob/main/README.md>

⁸<https://www.youtube.com/watch?v=vUj4VNXXC1c>;
<https://medium.com/the-nerd-circus/midjourney-prompt-weight-mastery-a-guide-to-advanced-prompt-crafting-b013c3bcade4>;
https://www.reddit.com/r/midjourney/comments/xdqbsx/whats_your_typical_flow/

user forums, such as Reddit and YouTube. These all suggest that prompts are also path-dependent.

4.2 Prompt Length Increases along Path

Users add words, parameters, and image inputs incrementally along the path. Figure 1 panel (a) plots the coefficients of $k^{th} Prompt$ in regression (1). The dependent variable is the number of words in prompt p written by user i in session a . The right-hand side consists of dummy variables indicating whether it is the k -th prompt in a session, along with the session fixed effects. If β increases with k , users are adding words to the prompt along the path.

In panel (a), the horizontal axis indicates whether it is the k -th prompt in a session, and the vertical axis is the coefficients and the 95% confidence intervals. I constrain the sample to be sessions with at least 15 prompts to avoid effects coming from the fact that longer search sequences have longer prompts. The figure shows that users are adding words to prompts along the path, at a decreasing rate. The effect is most salient in the first 12 prompts, and the trend slows down afterwards. Similar patterns arise if the dependent variable becomes the number of parameters or the number of image inputs.

$$\#Words_{isp} = \sum_k \beta_k \mathbb{1}(k^{th} Prompt_{isp}) + \mu_s^i + \epsilon_{isp} \quad (1)$$

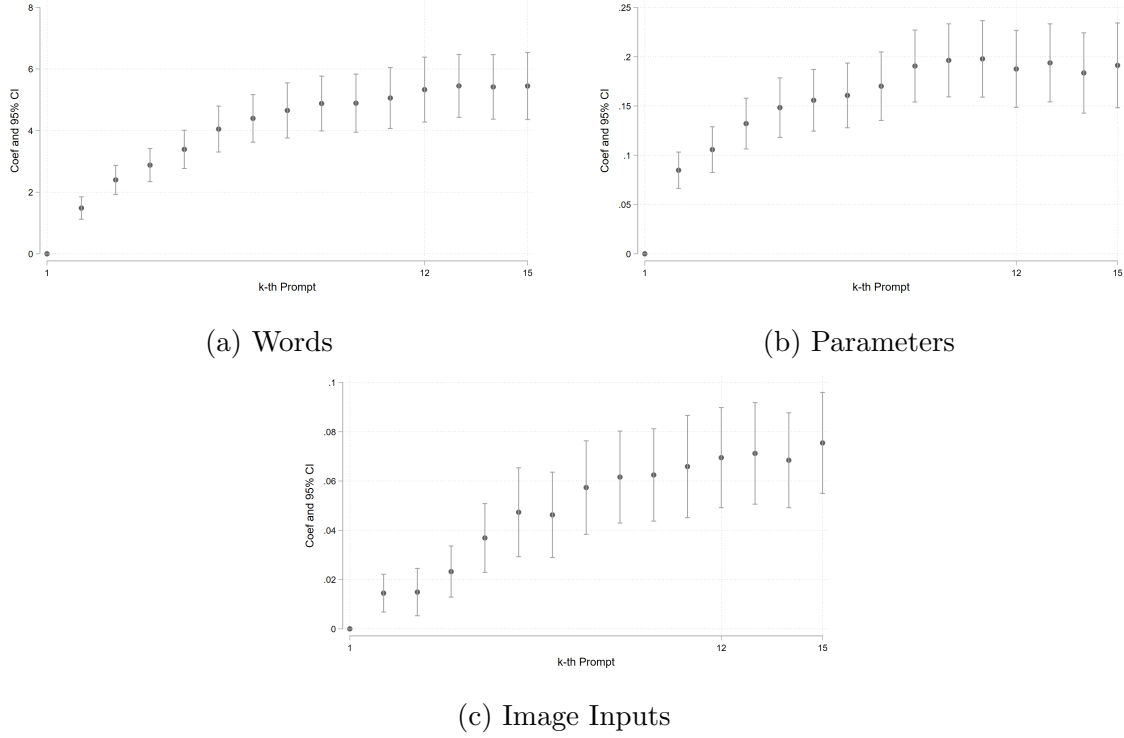


Figure 1: Prompt Length Increases Along The Path

Notes: Panel (a) in this figure shows the coefficients before $1(k^{th} Prompt)$ in regression (1). The horizontal axis indicates whether it is the k-th prompt in a session, and the vertical axis is the coefficients and the 95% confidence intervals. Standard errors are clustered at the user level. The reference point is the first prompt in a session. Panels (b) and (c) plot similar coefficients with the dependent variable being the number of parameters and the number of image inputs instead. I constrain the sample to be sessions with at least 15 prompts to avoid effects coming from the fact that longer search sequences have longer prompts. I have tried changing this number to 10 or 30, and the trends persist.

	(1)	(2)	(3)
Dep Var	<i>#Words</i>	<i>#Parameters</i>	<i>#Image Input</i>
$k^{th}Prompt$	0.336*** (0.037)	0.010*** (0.001)	0.005*** (0.0007)
Mean in 1 st Prompt	21.18	1.10	0.10
Session FE	Y	Y	Y
N	70,380	70,380	70,380
R^2	0.87	0.84	0.74

Notes: *** denotes significance at 1 percent, ** at 5 percent, and * at 10 percent. Standard errors are clustered at the user level. This table presents the estimates of regression (2). Here, I restrict the sample to be sessions with at least 15 prompts because I want to obtain the average effect of the prompt length growth in Figure 1.

Table 2: Prompt Length Grows Along The Path

Table 2 shows the average effects of the sample. It presents the results from regression (2). Although the effect seems small compared to the mean in the first prompt in a session, we need to keep in mind that the effects are diminishing along the path, and this is just the average effect. The effect in the first few prompts is not negligible, as shown in Figure 1.

$$\#Words_{isp} = \beta k^{th}Prompt_{isp} + \mu_s^i + \epsilon_{isp} \quad (2)$$

4.3 Prompts Become Increasingly Similar to the Final Prompt in a Session

Along the path, prompts are converging to the final prompt in a session. Figure 2 plots the coefficients before $\mathbb{1}(k^{th}Prompt_{iap})$ in regression (3). The dependent variable is the cosine similarity between prompt p written by user i in session a , and the final prompt in the same session. The cosine similarity is calculated based on prompt embeddings, which are vectors generated by large language models.⁹ Cosine similarity ranges from -1 to 1, with larger values indicating that two prompts are

⁹I employ both “sentence-transformers/all-MiniLM-L6-v2” in Python and “text-embedding-3-small” model from OpenAI. They give similar results.

more similar in meaning. The independent variables are dummies indicating whether this is the k -th prompt in the session, along with session fixed effects. Standard errors are clustered at the user level. If β is increasing with k , it implies that prompts are increasingly similar to the final prompt in a session.

$$\text{Cosine Similarity}(\text{Prompt}_{isp}, \text{Prompt}_{isT}) = \sum_k \beta_k \mathbb{1}(k^{th} \text{Prompt}_{isp}) + \mu_s^i + \epsilon_{isp} \quad (3)$$

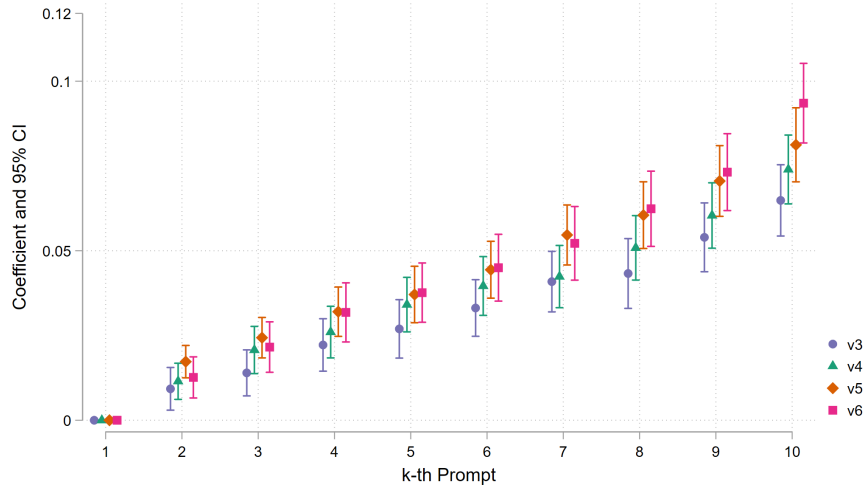


Figure 2: Convergence of Prompts Towards Final Prompt in Session

Notes: This figure plots coefficients before $\mathbb{1}(k^{th} \text{Prompt})$ in regression (3) and the 95% confidence intervals. I constrain the sample to be sessions with more than 10 prompts to avoid effects coming from longer search sequences converging faster. I also constrain the sample to be sessions with no more than 50 prompts because I want the first 10 prompts to be non-negligible in the search sequence. Imagine a search sequence with 500 prompts. The first 10 prompts only play a small part in the search sequence, and the effects can be noisy if the early convergence trend is very different from the late stages. I have tried different upper bounds and lower bounds, and the patterns persist. I also divide the sample by the Midjourney version they are using to show that users are converging at different rates when using different versions.

There are two trends worth noticing in Figure 2. Firstly, for a given Midjourney

version, β increases with k . This trend suggests that prompts are converging to the final prompt in a session along the path.

Secondly, prompts in later Midjourney versions are converging faster. For example, V4 has a larger β for every k in V3. This suggests that users are taking greater steps each time they search in later Midjourney versions.

Another way to verify that users take smaller steps in the prompt is to run regression (4). The dependent variable becomes the cosine similarity between prompt p and its previous prompt $p-1$. The independent variables are dummies representing the Midjourney version being used, along with user fixed effects. Figure 3 plots the coefficients of β . The results show that users take smaller steps in V3, but bigger steps in V4, V5, and V6.

$$\text{Cosine Similarity}(\text{Prompt}_{isp}, \text{Prompt}_{isp-1}) = \sum_v \beta_v \mathbf{1}(\text{Midjourney Version}_{isp} = v) + \mu_i + \epsilon_{isp} \quad (4)$$

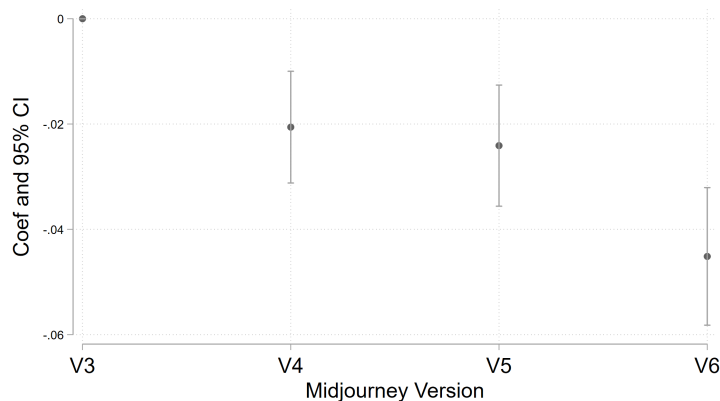


Figure 3: Larger Steps In Prompts With Newer AI

Notes: This figure plots coefficients before $\mathbf{1}(\text{Midjourney Version}_{iap} = v)$ in regression (4) and the 95% confidence intervals. I constrain the sample to be sessions with more than 10 prompts and no more than 50 prompts to be consistent with Figure 2. I have tried different upper bounds and lower bounds, and the patterns persist.

4.4 Users Take Larger Steps in Prompt Space Early in the Search Path

Users change words with higher weights in image outputs in early stages, and then they move to adjusting minor words. Words with higher weights are words that are closer to the beginning of the prompt. Online discussions show that users are aware of this rule. Some statements include “AI prioritizes the sequence of words in your prompt with the most important words being at the front of the prompt and descending from there, so be sure to organize the structure of your prompt by putting what you want to see most at the front of it”; “words 1-5 are very influential, ..., words 40+ are very likely to appear ignored”. Figure 4 shows an example. The words in the prompt are the same in the left and right panels, with the only difference being the order of words. In the left panel, “Pikachu” is at the beginning and “at park” is at the end, while in the right panel, the order flips. In response, AI returns images with a bigger Pikachu in the left panel and emphasizes the park on the right panel.

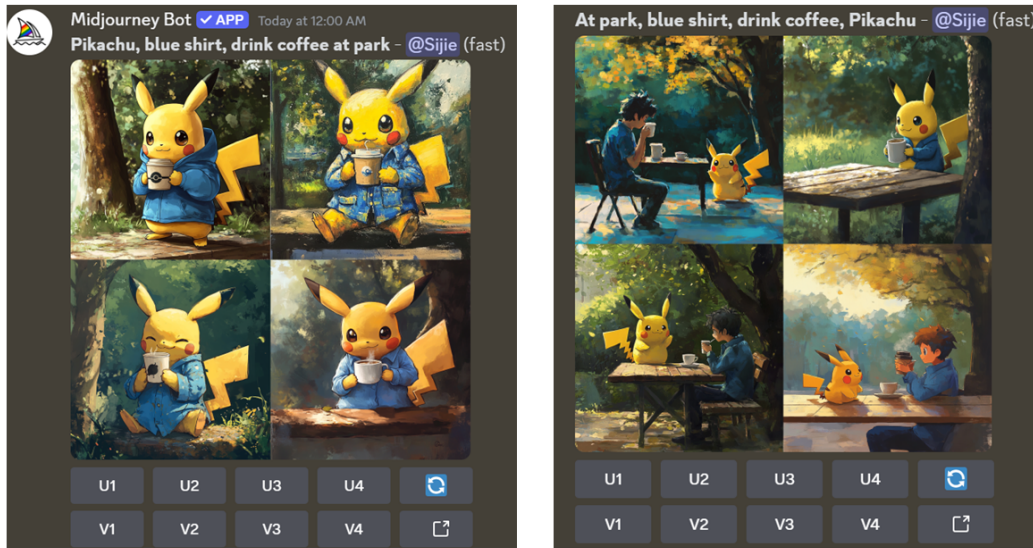


Figure 4: Words at the Beginning are More Important

The regression (5) examines whether the following prompt sequence is happening.

1. Pikachu, green shirt, drink coffee
2. Pikachu, **blue** shirt, drink coffee
3. Pikachu, **blue** shirt, drink **soda**

① ② ③ ④ ⑤

$$j^{th} Word\ Change_{isp} = \sum_k \beta_k \mathbb{1}(k^{th} Prompt_{isp}) + \mu_s^i + \epsilon_{isp} \quad (5)$$

The dependent variable indicates, compared to the most similar previous prompts, which is the word with the earliest position changed.¹⁰ Table 3 demonstrates what the data looks like. In the previous example, the first prompt is removed from the data because it does not have a previous prompt as a reference. In the second prompt, the earliest word that has changed compared to the first prompt is the second word “blue”. Hence, the dependent variable equals 2. Similarly, in the third prompt, the earliest word that has changed compared to the second prompt is the fifth word “soda”. The dependent variables are dummies indicating the prompt position in a session, along with session fixed effects. Standard errors are clustered at the user level. Figure 5 plots β and the 95% confidence intervals. An increasing β with k suggests that users change words closer to the beginning of the prompt first, and then move on to adjusting later words.

	Y	$\mathbb{1}(2^{ed} Prompt)$	$\mathbb{1}(3^{rd} Prompt)$	..
Pikachu, blue shirt, drink coffee	2	1	0	..
Pikachu, blue shirt, drink soda	5	0	1	..

Table 3: Sample Data For Regression (5)

A question here is whether this ordered word change happens horizontally or vertically. Denote a vertical change as adding words to the most similar previous

¹⁰By most similar, I mean the prompt with the most number of words at the beginning that are the same.

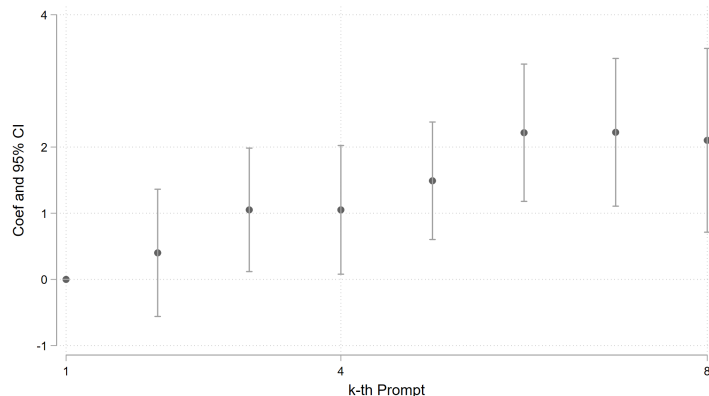


Figure 5: Change Early Words First

Notes: This figure plots coefficients before $1(k^{th} Prompt)$ in regression (5) and the 95% confidence intervals. I constrain the sample to be sessions with at least 8 prompts to avoid effects coming from longer search sequences changing words further away from the beginning of the prompt. I have tried different thresholds, and the trends persist.

prompt. In other words, one of the previous prompts is a strict subset of the current prompt. Denote a horizontal change as a change of words from the most similar previous prompt. The following example serves as an illustration.

Vertical

1. Pikachu
2. Pikachu, **blue shirt**
3. Pikachu, blue shirt, **drink coffee**
4. Pikachu, blue shirt, drink coffee **at café**

Horizontal

1. Pikachu, green shirt, drink coffee at café
2. Pikachu, **blue** shirt, drink coffee at café
3. Pikachu, **blue** shirt, drink **soda** at café
4. Pikachu, **blue** shirt, drink **soda** at **park**

I divide the sample into two subsamples. The first subsample only includes prompts that are a vertical change from the most similar previous prompt, and the second one includes all other prompts. I run regression (5) separately in both samples, and Figure 6 shows the results. The results suggest that the ordered word change happens both vertically and horizontally.

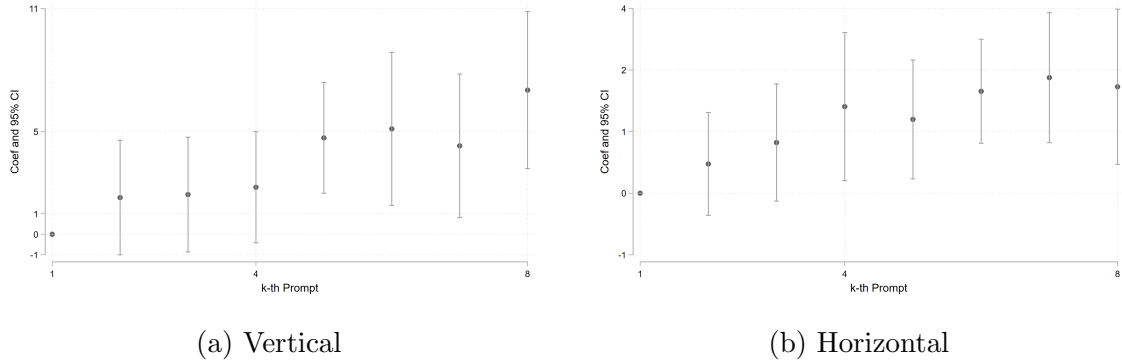


Figure 6: Ordered Word Change Happens Both Vertically and Horizontally

4.5 Discussion of Patterns Within a Creative Process

Consumer Search	User Writing Prompts
1. Query becomes longer / more specific along the search path.	1. Prompt length increases along the path.
2. State dependence: The Product searched currently has similar attributes to the product searched previously.	2. State dependence: Within a session, users adjust prompts step by step.
3. Products become increasingly similar to the final prompt in a path: As search progresses, products become increasingly similar to the product finally purchased.	3. Prompts become increasingly similar to the final prompt in a session: As search progresses, prompts become increasingly similar to the final prompt in a session.
4. Take larger steps in attribute space early in the search path: Consumers search a wider variety of products and take larger “steps” through attribute space early in the search path than later in the search path. For example, consumers explore a wider range of prices. And then later, they narrow it down to a smaller price range.	4. Take larger steps in prompt space early in the search path: Users adjust words with greater weights on image output early in the search path than later.

Table 4: Similarity Between Product Searching and Prompt Writing

These patterns mirror consumer search behavior documented in the literature as

described below. Table 4 summarizes the similarities between product searching and prompt writing.

In the consumer search literature (Bronnenberg et al., 2016; Hodgson and Lewis, 2025, Hirsch et al., 2020), researchers have shown that along the search path, the queries consumers submit to the search box become longer and more specific. For example, in Bronnenberg et al. (2016), the authors show that early queries tend to use generic terms such as “digital camera”, whereas later inquiries include specific models and brands like “Nikon” and “P520”. Similarly, Hirsch et al. (2020) documents an increase in query length as consumers refine their searches.

State dependence also arises in product search. Bronnenberg et al. (2016) and Hodgson and Lewis (2025) find that the probability for searching a Nikon camera increases by 0.250 if the brand was searched in the previous decile. These studies further show that the cameras that a consumer browses become increasingly similar to the camera they finally purchase along the path.

Additionally, consumers make larger attribute “steps” early in the search path. Bronnenberg et al. (2016) and Hodgson and Lewis (2025) find that early in their search, consumers explore a wide price range and diverse product attributes, before narrowing their focus in later stages.

In sum, although prompt writing and product searching seem different, they share some data patterns. One way to think about this is that a trained AI is a set of all possible images it can generate. Users are searching for the exact prompts that can “extract” the desired image. In the following section, I will characterize these patterns using the sequential search framework developed by Hodgson and Lewis (2025).

5 Data Processing

Since the set of possible prompts is all combinations of words in the English language, the prompt space is too large to handle. I follow the IO literature (Lancaster, 1966) and map prompts into attribute space to reduce dimensions. In addition, I focus on landscapes to better construct consideration sets of possible prompts. The

construction of consideration sets is discussed in Section 7.1. At this point, the focus is on sessions to create images with more than one prompt.

I split prompts into color, style, lighting, and a combined adjective/adverb category using the Midjourney online community classifications discussed in Section 3.2. An illustration of classified words is presented in Table C1.

Time	Text	X_{color}	$\mathbb{1}(Color)$	X_{style}	$\mathbb{1}(Style)$	...
0	Mountain, Black and White	-1	1	0	0	
1	Mountain, Black and White, Realistic	-1	1	0.9	1	
2	Mountain, Neon Color, Abstract	0.8	1	-0.7	1	
3	Mountain, Neon Color, Abstract	0.8	1	-0.7	1	

Table 5: Illustration of Prompt Attributes

To quantify these attributes, I embed the classified words with large language models and apply principal component analysis (PCA) to project each attribute’s embeddings onto a $[-1, 1]$ scale. An illustration of the attributes of prompts can be found in Table 5. Without specifying the direction of the numbers in each attribute, the embeddings and PCA automatically make colors that are more black and white into numbers closer to -1, and more colorful descriptions into numbers closer to 1. If the style is more realistic, the style attribute is closer to 1, compared to a style that is more abstract, with a number closer to -1.

I also include the dummy variables to indicate whether certain words are mentioned or not. For instance, if a prompt omits color terms, $\mathbb{1}(Color) = 0$. In this way, In this way, I am able to capture the difference between whether to specify certain attributes in the prompt, and the difference between different values of attributes.

6 Model Setup

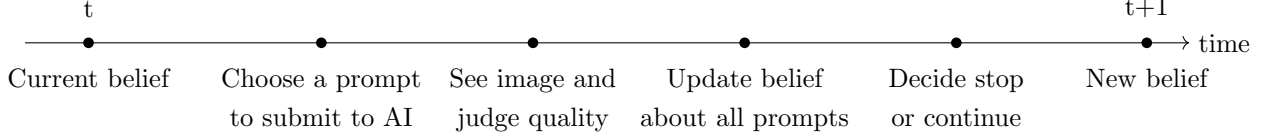


Figure 7: Timeline

Figure 7 describes the timeline of the model. In this model, an individual holds beliefs about the expected returns of the prompts in the consideration set at a given time t . The individual selects a prompt j , a combination of words, to submit to the AI. A search cost c_j is then incurred. The individual will see and judge the quality of the image, and update the belief about how good all prompts are. The search cost refers to the mental effort to construct prompts and judge the quality of an image; money paid to Midjourney, and wait time. Since the average monetary cost to Midjourney per prompt is less than 10 cents, and it takes, on average, less than 60 seconds for one generation, the search cost mainly refers to the mental effort. Based on active online discussions about prompt construction, it is usually mentally demanding to come up with the right prompt that generates satisfying results. After belief-updating, the individual will then decide whether to stop searching and end the session or to continue to search. If they decide to continue, the individual will decide which prompt to search next.

I assume that individuals search sequentially. In every period, they choose the prompt with the highest expected return based on their current belief. They stop searching when no remaining option offers sufficient expected returns to justify the search cost. The trade-off is between the option value of the next prompt and the search cost.

6.1 Individual's Problem

In a given prompt j , denote $\{X_k\}_j$ to be the attributes of the prompt. $\{X_k\}_j$ is eight dimensional, where $k \in \{color, style, lighting, adj/adv, \mathbb{1}(color), \mathbb{1}(style), \mathbb{1}(lighting), \mathbb{1}(adj/adv)\}$.

I treat the core subject of the prompt, such as the noun or object, as fixed within a session. This restriction keeps the model computationally tractable while focusing on the dimensions of prompts that are relevant for within-session search. Denote $\{\beta_k\}$ as the marginal return of attributes in individuals' *prior belief*. In session s , the utility of prompt j at time t is written in equation (6).

$$u_{sjt} = m_{st}(X_j) + \epsilon_{sjt} \quad (6)$$

where $\epsilon_{sjt} \sim N(0, \sigma_\epsilon)$ represents the idiosyncratic noise in each evaluation. This is because even if an individual submits the same prompt to AI multiple times, it will give different results, similar to other generative AI like ChatGPT. The AI model itself has randomness.

The function $m_{st}(X_j)$ is sampled from a Gaussian process with prior mean of

$$\mu_{s0}(X_j) = \sum_k X_{jk} \beta_k \quad (7)$$

and the covariance matrix

$$\kappa_{s0}(X_j, X_{j'}) = \lambda^2 \exp(-\|j - j'\|^2) \quad (8)$$

where $\|j - j'\|$ is the Euclidean distance of prompt j and j' embeddings, converted using OpenAI text-embedding-3-small model. The individual holds a prior mean μ_{s0} for the expected return of each prompt and a covariance matrix $\kappa_{s0}(X_j, X_{j'})$. If two prompts are similar in the embedding space, the covariance between them is high in prior belief. The randomness in $m_{st}(X_j)$ reflects the individual's uncertainty about how the AI interprets prompts without actually seeing the generated images.

At time t , after searches for prompt j , the individual observes the utility of the generated image u_{sjt} . The individual updates the beliefs of $m(X)$ function according to Bayes' rule, as illustrated in equation (9) and equation (10).

The intuition behind equation (9) is that observing the generated image for prompt j yields a realization of u_{sjt} . If u_{sjt} exceeds the current mean $\mu_{st}(X_j)$, the

posterior mean for all prompts increases, with larger increases for prompts whose embeddings are more similar to j . The intuition behind equation (10) is that sampling prompt j reduces the posterior variance for j most substantially, and also lowers the variance for other prompts. Uncertainty goes down because of the information gained. The reduction in uncertainty for each prompt is proportional to its similarity with j . Prompts that are more similar to j experience greater variance reduction.

$$\mu'(X) = \mu(X) + \frac{\kappa(X, X_j)(u_j - \mu(X_j))}{\kappa(X_j, X_j) + \sigma_\epsilon^2} \quad (9)$$

$$\kappa'(X, X') = \kappa(X, X') - \frac{\kappa(X, X_j)\kappa(X_j, X')}{\kappa(X_j, X_j) + \sigma_\epsilon} \quad (10)$$

Assume that at every period t , individuals consider the current choice as their final choice. In other words, in the individual's problem, the continuation value equals zero. This "think one step ahead" approach follows Hodgson and Lewis (2025). This assumption simplifies estimation.

A state in session s at time t comprises two components. The first one is $u_{st}^* = \max\{\hat{u}_{st}, 0\}$, which is the maximum between the highest utility among the past search results and the outside option utility normalized at zero. The second one is $f(u_{sjt})$, which represents the current belief of the utility of prompt j , with $u_{sjt} \sim N(\mu_{sjt}, \kappa_{st}^{jj} + \sigma_\epsilon^2)$.

Denote the search cost of prompt j as c_{sjt} . Assume $c_{sjt} = c + \zeta_{sjt}$, $\zeta_{sjt} \sim \text{EVT1}$ distribution, following Hodgson and Lewis (2025). The logit term ζ_{sjt} is drawn independently across s , t and j . This logit error assumption simplifies subsequent computation and captures heterogeneous search costs across sessions, time, and prompts. Note that search cost includes the mental effort required to construct prompts and assess the quality of generated images, the money paid to Midjourney, and the wait time. The heterogeneity primarily comes from differences in mental effort.

Given state variables $\{u_{st}^*, f(u_{sjt})\}$, the individual's problem can be written as:

$$\max \left\{ \underbrace{u_{st}^*}_{\text{stop}}, \underbrace{u_{st}^* + \max_{\hat{j}} \left\{ \int_{u_{st}^*}^{\infty} (u_{s\hat{j}t} - u_{st}^*) f(u_{s\hat{j}t}) du_{s\hat{j}t} - c_{s\hat{j}t} \right\}}_{\text{continue searching}} \right\} \quad (11)$$

If the user stops searching, the individual secures the current best utility u_{st}^* with certainty. If the user continues, the individual will at least collect the current best utility and select prompt j to maximize the expected premium, in addition to the search cost. Similar to Weitzman (1979) and Hodgson and Lewis (2025), a prompt that has a more dispersed distribution on the right end is more favored.

A distinction that I make from Hodgson and Lewis (2025) is that prompts can be resubmitted to better fit the setting. Even if the same prompt is submitted to AI multiple times, it gives a random image each time. The scale of σ_ϵ will determine how often users revisit the same prompt. If AI randomness σ_ϵ is large, we will see more revisits in the data. In the data for estimation, 14.16% of the prompts are revisit prompts, which are prompts that show up more than once in a session.

6.1.1 Discussions on Interpretations of Learning

There are two interpretations of learning in this context. Anecdotal evidence suggests that some users know what they want from the beginning. They have a clear picture in mind and adjust prompts to gradually converge on this target image. For example, an Instagram influencer known for using Midjourney to create images mentioned in a conversation with me: “Most of the time, I know exactly what I want to create. ... I know how to adjust my prompts to get closer to my original vision — not always perfectly, but I always start with a clear concept.”

Another group of users begins with only a vague idea of what they want. As they prompt, they gradually form and refine their desired outcome. This behavior is not unique to AI-assisted art and already existed before such tools were available. For instance, professional illustrator Christoph Niemann writes in the New York Times Magazine: “Creating art is a nonlinear process. I start with a rough goal. But then I head into dead ends and get lost or stuck. The secret to my process is to be on high

alert in this deep jungle for unexpected twists and turns, because this is where a new idea is born.” In the Midjourney chat channel, there is a “micro-polls” sub-channel initiated by Midjourney’s CEO, David Holz, to understand users and later construct larger questionnaires. Two questions are particularly relevant (see Figure C1 in the appendix):

1. What best describes your usage of Midjourney?
 - (a) I like to focus on a single goal (489 users)
 - (b) I like to explore many goals/ideas simultaneously (1,639 users)
2. What best describes your usage of Midjourney?
 - (a) I am goal-focused (1,104 users)
 - (b) I mostly wander around (1,057 users)

Economically, we can interpret these two approaches as follows. First, some users start with a well-defined goal and learn which prompts bring them as close as possible to that goal. Second, other users learn about their own preferences as they interact with Midjourney. While I cannot empirically distinguish these two cases in my data, Hodgson and Lewis, 2025 offers similar interpretations in a consumer search model. In the context of consumers searching for products to purchase, one interpretation is that consumers have a clear idea of the product they want at the beginning and search across stores for the digital camera that best matches this target. The alternative interpretation is that consumers do not fully understand their own preferences and, as they search, they learn about those preferences. The notion of learning in this paper should be interpreted in exactly these two ways.

7 Identification and Estimation

Given the individual’s problem, the probability of choosing prompt j can be written as follows. The idea is that if prompt j has a higher expected return given the current

state, it will be chosen with a higher probability.

$$P(j|u^*, f) = \frac{\exp \left[u^* + \int_{u^*}^{\infty} (u_j - u^*) f(u_j) du_j - c \right]}{\exp(u^*) + \sum_{\hat{j} \in J} \left[u^* + \int_{u^*}^{\infty} (u_{\hat{j}} - u^*) f(u_{\hat{j}}) du_{\hat{j}} - c \right]} \quad (12)$$

The likelihood of observing prompt sequences of session s in the data can be written as:

$$L_s = \int \prod_{t=0}^{T-1} P_s(j|u_t^*, f_t) \times P_i(0|u_T^*, f_T) dF(\epsilon) \quad (13)$$

The likelihood of observing all sessions in the data is then:

$$L = \prod_s L_s \quad (14)$$

I estimate the model based using MLE. The parameters to be identified are:

$$\psi = \left(\underbrace{\vec{\beta}}_{\text{Prior Mean}}, \underbrace{\sigma_{\epsilon}}_{\text{AI randomness}}, \underbrace{\lambda}_{\text{learning parameter}}, \underbrace{c}_{\text{search cost}} \right) \quad (15)$$

Table 6 presents the identification intuition of the parameters. $\vec{\beta}$ is the expected return of attributes in prior belief in equation (7). It will be identified by the first prompt in a session. λ is the coefficient before the covariance matrix in equation (8). If λ increases, the relative importance of the covariance between prompts increases. When prompt j is searched, the variance of prompt j and similar prompts decreases significantly if λ is large. This is because if the covariance between two prompts is high, when the individual obtains information from one prompt, the uncertainty of the related prompt decreases significantly. λ is identified by how far the individual jumps away from prompt j after the individual searches for it.

The parameter σ_{ϵ} governs the AI randomness. Suppose that every time the same prompt is submitted to AI, AI gives the same image. Users will not have incentives to revisit the same prompt since it is costly. Hence, if AI is not random at all, we should expect to see no revisit behaviors in the data. If, instead, AI gives very different images even if the same prompt is resubmitted to AI, we should expect to see more revisit behaviors in the data. The parameter σ_{ϵ} will be identified by the

frequency of resubmission of the same prompt in the data. The parameter c is the average search cost, which will be identified by the average search length. If the search cost is high, the average search length should be short.

Parameters	Interpretation	Identification Intuition
$\vec{\beta}$	Prior belief of attributes	Words in first search
λ	Covariance Scale	Jump distance after search prompt j
σ_ϵ	AI Randomness	Frequency of resubmitting the same prompt
c	Search Cost	Search length

Table 6: Identification Intuition

7.1 Construct Consideration Set

Before estimating the model, I need to define the set of prompt choices available to the user at each step. In the model, the user chooses the next prompt from a consideration set. This set represents the feasible prompt alternatives that the user may consider after observing the image generated by the previous prompt. Constructing this set is necessary because the full prompt space is extremely large: in principle, a user could write any combination of words in the English language. In the choice probability in equation (12), all possible prompt choices enter the denominator. If all possible prompts were included as alternatives, this denominator would contain millions of options, making the probability of choosing any observed prompt mechanically close to zero and weakening identification. I therefore approximate the user’s feasible choice set using observed prompt transitions in the data.

In practice, users do not consider all combinations of words in the English language at each step. The path-dependent clusters of prompts documented in Section 3.1 imply that users restrict their choices to a local consideration set, prompts similar to the previous one, and consistent with the idea of an artwork.

I construct consideration sets under two constraints. First, I restrict estimation

to landscape sessions, defined as sessions whose core subject is scenery, such as mountains, lakes, forests, or other natural environments. Appendix B.5 describes the classification procedure in detail. Estimates for portrait sessions, defined as sessions whose core subject is a person, character or face, yield similar results, which can be found in Table D2 in the appendix.

Second, I include in the consideration set for session s only prompts that (1) exist in the data; (2) lie within a distance threshold from any prompt in session s . To find this distance threshold, I compute embedding distances for all consecutive prompt pairs in landscape sessions. Figure C2 in the appendix shows that over 95% of consecutive pairs lie within a distance of 4.7. I adopt this value as the threshold, yielding an average consideration set size of 21.85 prompts per session.

Another way to construct consideration sets is to identify the N nearest prompts in the data for each prompt in the session. To make estimation results comparable, I choose $N = 14$ to match the average set size from the distance-based method. The resulting average set size is 21.35. I estimate the model under both consideration set definitions.

7.2 Estimation Results

Table 7 reports the estimation results of the parameters. The complete results with β , the marginal return of attributes in individuals' prior belief, are presented in Table D1 in the appendix. Because the raw scales lack direct economic interpretation, their implications should be interpreted via counterfactual simulations. The results from the two consideration set definitions are very similar to each other. The main takeaway here is that the covariance matrix scale λ is significantly greater than zero, indicating that covariance plays a crucial role in belief updating. Users indeed adjust prompts based on judgment of the previous results.

Consideration Set	(1) Prompts Within Distance	(2) N Nearest Prompts
λ (Covariance Scale)	81.82 (1.64)	80.97 (1.56)
σ_ϵ (AI Randomness)	13.44 (0.32)	14.24 (0.33)
c (Search Cost)	6.43 (0.08)	6.66 (0.08)

Notes: Standard errors in brackets, computed using the observed Fisher information.

Table 7: Model Estimates

8 Counterfactual

The key element of human adaptation to output is the individual’s ability to *judge* the quality of generated images and adjust subsequent prompts accordingly. This impacts the direction of search, and hence the production process. To understand the role of such adaptation, I conduct a counterfactual analysis where I eliminate this kind of judgment by setting $cov(j, j') = 0, \forall j \neq j'$.

In the most extreme case, all judgment is removed. In that case, individuals are submitting prompts randomly at each period. The mathematician Émile Borel once used a metaphor of the infinite monkey: Imagine a monkey hitting keys independently and randomly on a typewriter for an infinite amount of time. Almost surely, it will type any given text, including the complete works of William Shakespeare. In the context of image generation, if we iterate over every combination of pixels on a canvas, we will almost surely obtain some masterpieces. However, it is not an efficient means of production. We still need humans to go through all these images and select the masterpieces based on their judgment.

In this counterfactual, I am making the individuals smarter than the infinite monkey, in the sense that they can submit a list of prompts to the AI based on their

prior beliefs. But the individuals are less adaptive than individuals in the data, in the sense that they are not allowed to adjust prompts based on judgment of previous output.

The prompts with a higher expected return in prior beliefs are ranked higher in the predetermined list. The AI will proceed with the order of the list and generate images. However, unlike the reality setting, the individual is not allowed to change the order of the list based on the judgment of the images. The list is infinitely long. The counterfactual exercise is designed to ask: how many more prompts are needed to reach the same final prompt in a session as in the data?

Table 8 presents the results of the counterfactual analysis. Without human adaptation directing the prompt-searching behavior, it takes around three times more prompts per session to achieve the same outcome. The results suggest that human adaptation plays a crucial role in directing search. When an individual sees an unsatisfying image, the prompt is adjusted to move away from the previous one. This adaptation increases production efficiency by around three times. This number highlights the importance of human adaptation to the creative process.

	(1)	(2)
Consideration Set	Prompts Within Distance	N Nearest Prompts
#Prompts per Session Increases	313%	284%

Notes: The table reports, for each consideration-set definition, how much longer sessions would be in the counterfactual where users cannot adapt prompts to previously observed images. The outcome is the percentage increase in the number of prompts needed to reach the same final prompt as in the data. Column (1) defines the consideration set as all prompts within an embedding-distance threshold from the session’s prompts; column (2) defines it as the N nearest prompts in the data. In both cases, removing adaptation requires roughly three times more prompts.

Table 8: Counterfactual Results

9 Conclusion

Using data from Midjourney, I show that creative production with generative AI is not a single step, but an iterative process in which users learn from previous outputs and adjust their prompts accordingly. Within a creative process, prompts are path-dependent. Prompts become longer and increasingly similar to the final prompt along the path. Users change words with higher weight first and then move to details. The patterns are consistent with a process of costly sequential search with learning.

Motivated by the reduced form evidence, I build a sequential search model to capture the prompt patterns. In the model, individuals search sequentially for the prompts that are correlated. When a prompt is searched, the individual sees the generated image, evaluates the quality, and adjusts the next prompt accordingly. A counterfactual simulation that disables output assessment reveals that each session requires 313% more prompts, on average, to reach the same final outcome. This result shows that given the initial idea, human judgment contributes to creative production through the ongoing process of evaluating outputs, learning from them, and redirecting the search. Even when generative AI can produce an image within seconds, human effort remains central because users guide the model toward desired outcomes through iterative refinement.

More broadly, this paper opens the black box of the creative process. In many settings, researchers observe only the initial inputs and final outputs of creative work. By contrast, the Midjourney setting allows observation of inputs and outputs of intermediate steps, making it possible to study how humans create step-by-step. It also offers an infrastructure for future research on creative work.

Appendix

A Screenshots About User Composition

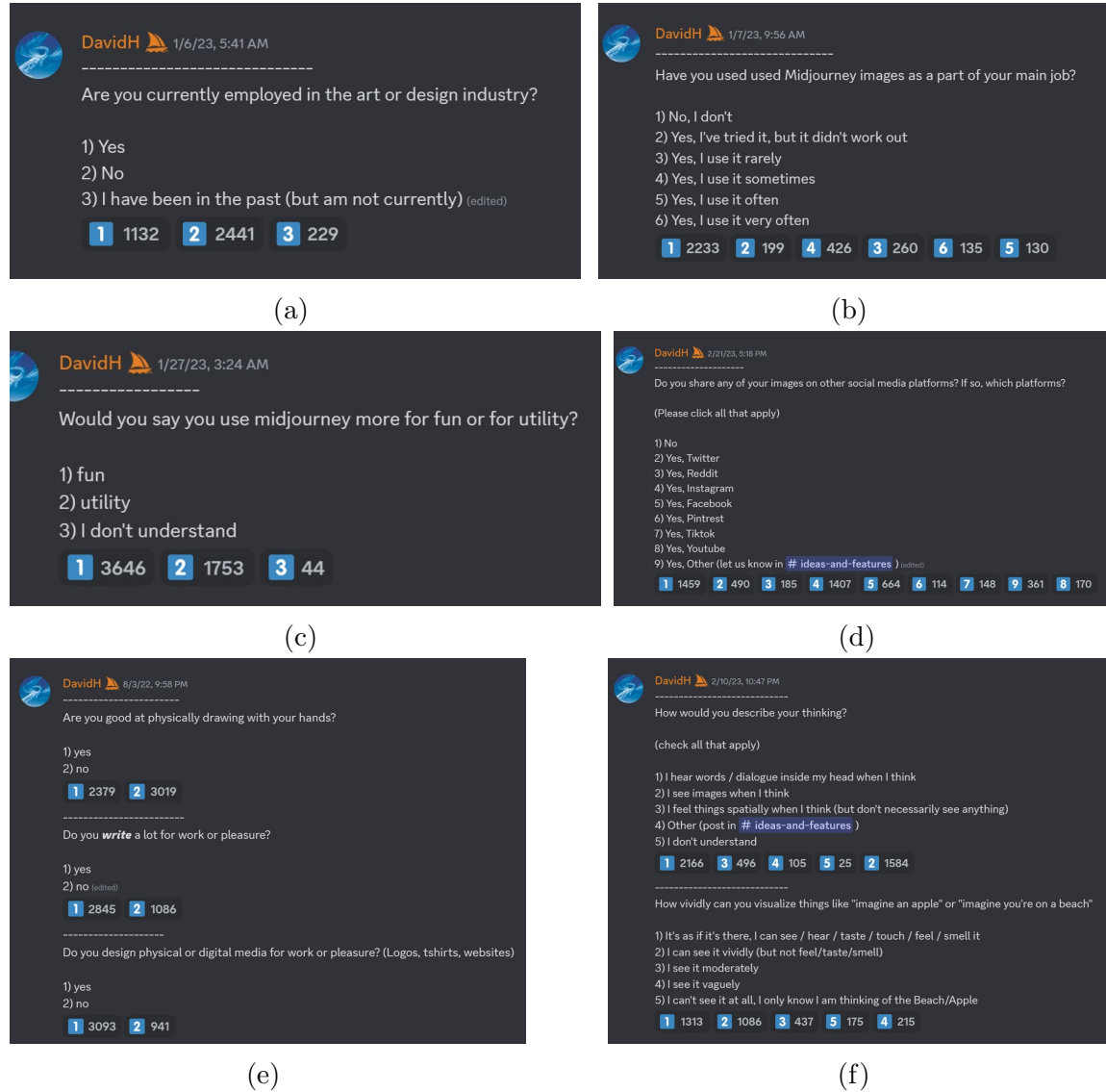
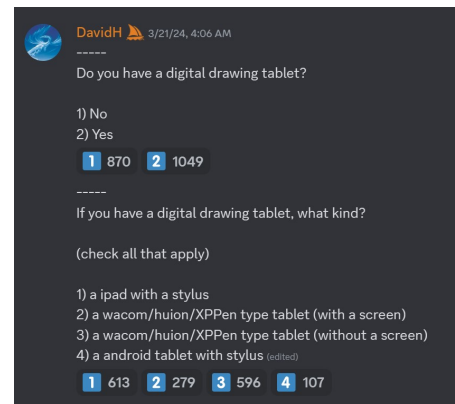


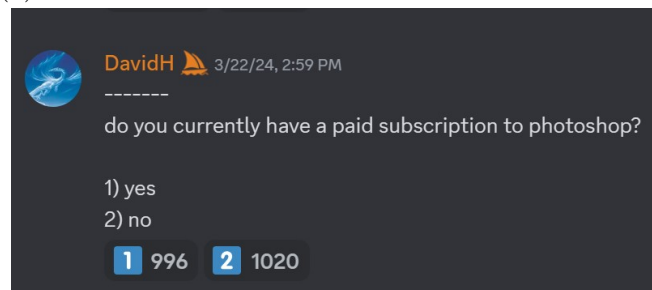
Figure A1: "Mini Poll" Screenshots 1



(a)



(b)



(c)

Figure A2: “Mini Poll” Screenshots 2

midjourney.com Compare

Audience Interests

Oct, 2023 - Dec, 2024 (Custom) United States

	Domain (9,561)	Industry	Global Rank	Relevance Score	Cross-Visitation	PoP change	AdSense
1	app.runwayml.com	Games > Video Games Consoles and Acco...	-	100	3.22%	-	-
2	app.leonardo.ai	Computers Electronics and Technology > C...	-	98	3.15%	-	-
3	elevenlabs.io	Computers Electronics and Technology > P...	#3,890	90	2.95%	-	-
4	runwayml.com	Computers Electronics and Technology > G...	#6,988	89	2.25%	-	-
5	printify.com	Business and Consumer Services > Busine...	#5,806	86	2.74%	-	-
6	labs.openai.com	Games > Video Games Consoles and Acco...	-	80	2.34%	-	-
7	ideogram.ai	Computers Electronics and Technology > P...	#4,336	79	2.11%	-	-
8	freeml.com	Arts and Entertainment > Arts and Entertai...	#225	76	2.95%	-	-
9	openai.com	Computers Electronics and Technology > C...	#88	75	7.61%	-	-
10	huggingface.co	Computers Electronics and Technology > C...	#2,379	74	2.67%	-	-
11	leonardo.ai	Computers Electronics and Technology > C...	#4,269	73	1.73%	-	-
12	canva.com	Computers Electronics and Technology > G...	#39	72	15.28%	-	-
13	claude.ai	Computers Electronics and Technology > P...	#797	71	4.66%	-	-
14	platform.openai.com	Computers Electronics and Technology > P...	-	67	3.20%	-	-
15	lumalabs.ai	Computers Electronics and Technology > P...	#10,244	66	1.60%	-	-
16	checkout.stripe.com	Computers Electronics and Technology > C...	-	66	7.18%	-	-
17	civitai.com	Computers Electronics and Technology > S...	#2,340	65	1.93%	-	-
18	behance.net	Arts and Entertainment > Visual Arts and D...	#1,314	65	2.07%	-	-
19	openart.ai	Computers Electronics and Technology > C...	#9,849	64	1.61%	-	-
20	freeml.com	Computers Electronics and Technology	#894	63	3.27%	-	-
21	auth0.openai.com	Computers Electronics and Technology > P...	-	63	6.00%	-	-
22	freeml.com	Computers Electronics and Technology > G...	-	62	1.46%	-	-

(a)

docs.midjourney.com Compare

Audience Interests

Oct, 2023 - Dec, 2024 (Custom) United States

	Domain (2,129)	Industry	Global Rank	Relevance Score	Cross-Visitation	PoP change	AdSense
1	midlibrary.io	Computers Electronics and Technology > S...	#202,470	100	4.84%	-	-
2	app.runwayml.com	Games > Video Games Consoles and Acco...	-	97	5.87%	-	-
3	checkout.stripe.com	Computers Electronics and Technology > C...	-	95	13.37%	-	-
4	runwayml.com	Computers Electronics and Technology > G...	#6,988	94	5.06%	-	-
5	app.leonardo.ai	Computers Electronics and Technology > C...	-	89	5.43%	-	-
6	openart.ai	Computers Electronics and Technology > C...	#9,849	85	4.81%	-	-
7	bootcamp.udesign.cc	Computers Electronics and Technology > C...	-	84	3.46%	-	-
8	prompthero.com	Computers Electronics and Technology > G...	#43,763	83	3.93%	-	-
9	ideogram.ai	Computers Electronics and Technology > P...	#4,336	81	4.86%	-	-
10	stability.ai	Computers Electronics and Technology > P...	#51,585	79	3.83%	-	-
11	leonardo.ai	Computers Electronics and Technology > C...	#4,269	76	3.42%	-	-
12	printify.com	Business and Consumer Services > Busine...	#5,806	76	5.68%	-	-
13	promptbase.com	Computers Electronics and Technology > C...	#119,644	75	3.36%	-	-
14	elevenlabs.io	Computers Electronics and Technology > P...	#3,890	72	5.14%	-	-
15	help.openai.com	Computers Electronics and Technology > P...	-	70	4.27%	-	-
16	labs.openai.com	Games > Video Games Consoles and Acco...	-	70	3.20%	-	-
17	civitai.com	Computers Electronics and Technology > S...	#2,340	69	5.02%	-	-
18	pika.art	Computers Electronics and Technology > G...	#20,537	67	3.74%	-	-
19	zapier.com	Computers Electronics and Technology > P...	#6,214	67	5.24%	-	-
20	creator.nightcafe.studio	Computers Electronics and Technology > G...	-	67	3.68%	-	-
21	support.discord.com	Games > Video Games Consoles and Acco...	-	66	7.54%	-	-
22	craiyon.com	Games > Video Games Consoles and Acco...	#22,589	65	2.85%	-	-

(b)

Figure A3: SimilarWeb Screenshots

B Data Appendix

B.1 Data Collection

Data collection started in February 2, 2025 and ended in February 8, 2025. I collected messages from channels called “general 1”, “general 2”, ..., “general 20”.

B.2 Clean Errors And Link Type 19 Messages to Type 0 Messages

The data cleaning process follows the following steps.

1. Read all messages.
2. Delete messages whose type is not 0 or 19. Type 0 messages are those prompts submitted by users. Type 19 messages are the result of clicked buttons. I also delete those that are not sent by the Midjourney Bot and those with errors.
3. For each message of type 19, identify the *message_reference* to determine which message the user interacted with to trigger the current message. By doing so, I can trace backward until a type 0 message is found. For a given type 0 message, if there is more than one user involved along the path, meaning that more than one user clicks on the buttons, I remove this type 0 message and all the subsequent messages.
4. Break “content” in each message into text inputs, parameters, image inputs, and mode.
5. For each type 0 message, if at least one image input comes from the generated output of another user, remove the path (keep the original user’s path).

B.3 Clustering Prompts Into Sessions

This section describes the details of clustering prompts into sessions. Table B1 presents the distribution of session time spans.

1. For text inputs in each type 0 message, remove stop words and “magic words”.
 - Stop words are words like “a”, “an”, “the”, “here”. This is a terminology usually employed by computer scientists who study natural language processing. I am using the list from the “nltk.corpus” package in Python.
2. For a given user, collect all type 0 messages and find N cleaned text inputs. Calculate a $N \times N$ Jaccard similarity matrix. The Jaccard similarity score is between 0 and 1, with a larger number meaning texts being more similar.
 - For each type 0 message, if at least one of the image inputs comes from the generated output of the **same user**, assign a similarity score of one to guarantee that they are in the same session.
3. For prompt pairs in which at least one of them does not have text input, look at the image inputs. If at least one of the image inputs is the same, assign a similarity score of one to guarantee that they are in the same session.
4. Use DBSCAN to cluster type 0 messages within a user. Assign the threshold to be 0.2.
5. For each session, find the first action by timestamp and remove it if it is within the last 24 hours of the data collection period. This is to remove those sessions that have just started, but I did not collect the complete path due to a fixed data collection start time.

Session Time Span	%
0–1 h	90.62
1–6 h	1.30
6–12 h	0.37
12–24 h	0.55
24–48 h	0.47
48–72 h	0.25
3–7 days	0.56
7–30 days	1.11
> 30 days	4.77

Table B1: Distribution of Session Time Span

B.4 Word List For Classifications

Below are some examples of the word classifications. The complete list will be available in the online appendix.

1. *detailed words*: detail, detailed, high-detailed, ultra-detailed, hyper-detailed, extremely-detailed
2. *realistic words*: realistic, hyperrealistic, super realistic, photorealistic, photorealism
3. *rendering engines*: rendering engine, unreal engine, Cinema4D, C4D, Arnold Render, Octane Render
4. *colors*: vibrant colors, black, white, emerald, olive-green, vivid-maroon, cyan, neon colors, light yellow
5. *stop words*: in, on, just, me, as, is, are, am, there, here, had, while
6. *photography words*: fisheye lens effect, 135mm lens, white balance, shutter speed, stop motion

7. *lighting*: tyndall effect, dim lighting, glowing radioactivity, studio light, dj lighting

8. *style*: post-impressionism, neo-pop art, street art, futurism, cubist, dadaism

B.5 Clustering Sessions into Topics

I classify sessions into topics using a popular language-processing Python package called BERTopic. It is widely used in classifying documents into topics using clustering.

For prompts in a single session, I append the text inputs together. And then I remove the color words and stop words to improve clustering accuracy. I then cluster sessions into 100 topics using BERTopic, with the K-Means clustering method. Below are some examples of topics, with corresponding keywords, and some sampled prompts in each topic.

1. Landscapes

Keywords: mountains, lake, sky, landscape, desert, river, sunset, water

Sampled Prompts:

- “new Mexico landscape, clear blue sky, rocky mountains, rainclouds, dramatic, oil painting, detailed, Albert bierstadt style”
- “dramatic cloud formation coming from behind a mountain, clear blue sky, midday, bright, southwest, colorful, awe-inspiring, detailed, oil painting”

2. Cute Animals

Keywords: cat, dog, cute, kitten, adorable, puppy, fur, bunny

Sampled Prompts:

- “super cute black french bulldog + big expressive eyes + Pixar + ghibli + extremely detailed + detailed reflections + bold highlights + photo-realistic + hdr + octane render + 8K + soft light + twinkling fire + smoke”

- “white cat, roller skating, playing keytar, wearing sunglasses, playing keytar”

3. Architectures and Room Designs

Keywords: room, interior, house, modern, design, walls, furniture, architecture, wall

Sampled Prompts:

- “studio with architects designing models of future cities”
- “luxurious comfortable biopunk lounge inside a huge tropical greenhouse, volumetric lighting, hyperdetailed, 8k octane render”

4. Portraits

Keywords: hair, eyes, face, woman, portrait

Sampled Prompts:

- “tall 50 year old female police officer with Curly blonde hair and wearing black framed reading glasses, realistic, Norman Rockwell style, 10k”
- “long red hair, beautiful woman in green dress”

C Additional Tables and Figures

Time	Text	Color	Style	...
0	Mountain, Black and White	Black and White	-	
1	Mountain, Black and White, Realistic	Black and White	Realistic	
2	Mountain, Neon Color, Abstract	Neon Color	Abstract	
3	Mountain, Neon Color, Abstract	Neon Color	Abstract	

Table C1: Illustration of Splitting Texts

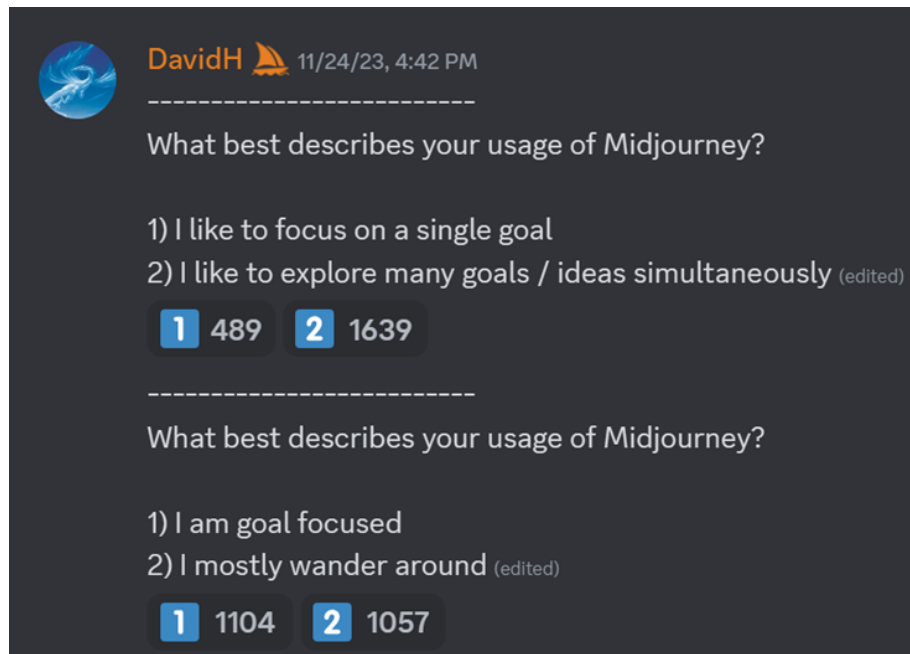


Figure C1: “Mini Poll” Screenshot

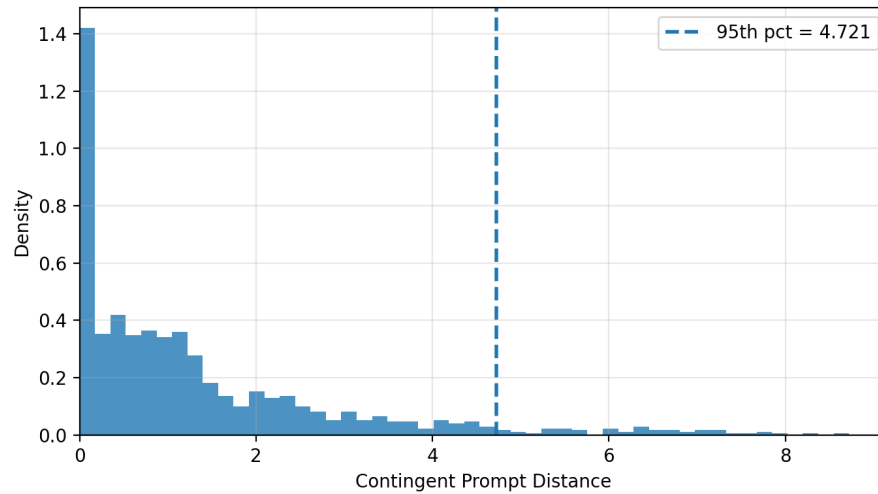


Figure C2: Contingent Prompt Distance Distribution

D Complete Model Estimates

Consideration Set	(1) Prompts Within Distance	(2) N Nearest Prompts
λ (Covariance Scale)	81.82 (1.64)	80.97 (1.56)
σ_ϵ (AI Randomness)	13.44 (0.32)	14.24 (0.33)
c (Search Cost)	6.43 (0.08)	6.66 (0.08)
β_{color}	0.18 (0.29)	-0.12 (0.29)
β_{style}	-0.70 (0.22)	-0.56 (0.22)
$\beta_{lighting}$	0.95 (0.44)	0.70 (0.48)
$\beta_{adj/adv}$	0.35 (0.18)	0.54 (0.17)
$\mathbb{1}(color)$	-0.75 (0.21)	-1.05 (0.19)
$\mathbb{1}(style)$	-1.48 (0.21)	-1.80 (0.21)
$\mathbb{1}(lighting)$	-1.34 (0.31)	-1.32 (0.33)
$\mathbb{1}(adj/adv)$	-1.01 (0.25)	-1.54 (0.22)

Notes: Standard errors in brackets, computed using the observed Fisher information.

Table D1: Model Estimates (Complete)

Consideration Set	(1) Prompts Within Distance	(2) N Nearest Prompts
λ (Covariance Scale)	99.43 (2.77)	98.87 (2.77)
σ_ϵ (AI Randomness)	17.05 (0.59)	18.31 (0.59)
c (Search Cost)	8.13 (0.19)	8.49 (0.19)
β_{color}	-0.41 (0.29)	-0.21 (0.29)
β_{style}	-2.21 (0.25)	-2.40 (0.25)
$\beta_{body\ part}$	0.73 (0.21)	0.87 (0.21)
$\beta_{adj/adv}$	-0.14 (0.28)	-0.10 (0.28)
$\mathbb{1}(color)$	-0.34 (0.25)	-0.72 (0.25)
$\mathbb{1}(style)$	-2.28 (0.36)	-2.88 (0.36)
$\mathbb{1}(body\ part)$	-1.87 (0.24)	-2.15 (0.24)
$\mathbb{1}(adj/adv)$	-1.98 (0.44)	-2.44 (0.44)

Notes: Standard errors in brackets, computed using the observed Fisher information.

Table D2: Model Estimates (Portrait)

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